

Feedlot Margins, Cattle Prices and Beef-Exposed Equities: Two Pre-Registered Tests

Gator Quant Hacks 2026 · Systematic Trading Track · [team names] · Research note; holdout evaluated once on 2026-10-04

Summary. We pre-registered two hypotheses linking cattle markets to tradable returns, tested each frozen primary specification on development data (2010–2024-09), and then evaluated both once on a two-year holdout (2024-10-01 to 2026-10-02).

- **H1 (main study):** deeply negative feedlot margins did not predict fewer cattle placements (P1: $\beta = +0.0020$ per s.d., Newey-West $t = 0.42$), and the primary trading rule earned a net Sharpe ratio of -0.08 (P2: mean monthly net return -0.045% , $t = -0.34$). By the pre-registered decision table, H1 is **rejected** for this specification.
- **H2 (follow-up):** Tyson and Texas Roadhouse, hedged with their sector ETFs, showed no same-month cattle exposure (Q1: $t = +0.10$) and no lagged reaction (Q2: $t = +0.88$); the primary rule earned a net Sharpe ratio of -0.29 . H2 is **rejected** for this specification.
- Across the 25 cattle-project trials, full-sample development net Sharpe ratios range from -0.42 to $+0.21$; no monthly-return t -statistic exceeds 0.86 .
- **Holdout:** H1's primary earned a net Sharpe ratio of $+0.63$ ($t = 0.97$), meeting P2's holdout condition, but P2 also required in-sample significance and P1 had already failed, so the decision table still rejects H1. H2's primary earned -0.15 and fails Q3.

These results show no support for the tested specifications. They are not evidence that every cattle-related strategy is unprofitable.

1. Economic mechanisms and hypotheses

H1: feeding margins and Live Cattle futures. A feedlot buys an 800 lb feeder steer, feeds it about 64 bushels of corn and sells it at 1,400 lb roughly five months later. The projected margin is $M_t = 1,400 \times LE_t - 800 \times FC_t - 64.29 \times C_t - \350 , using CME settlements of a Live Cattle contract expiring at least 150 days out, the nearest Feeder Cattle contract with 30 days left and a corn contract at least 75 days out. Two channels could make low margins bullish for deferred cattle futures: low margins cut placements, so fewer fed cattle reach market five to six months later; and hedging feedlots pay speculators a premium to carry price risk (Keynes 1930; de Roon, Nijman & Veld 2000). The signal is z_t , the margin's distance from its mean over the 36 prior month-ends in standard deviations (the \$350 non-corn cost cancels). Tests: **P1** low z predicts fewer placements over months $t+1..t+3$; **P2** the primary rule (Variant A: $w = \text{clip}(-z/2, -1, 1) \times 10\% / \hat{\sigma}$, $|w| \leq 2$, long or short the deferred contract) earns a positive in-sample net return with $t > 2$ and a positive holdout Sharpe; **P3** the kill test asks whether z predicts the realized spot change but not the futures return, meaning the curve already prices the supply response. Variants B (the full margin spread) and C (A filtered by commercial hedging pressure) are pre-registered secondary rules.

H2: cattle prices and beef-buying equities. Cattle are a major input for Tyson (beef segment) and Texas Roadhouse (largest commodity cost). If equity investors underreact to cost news that originates in a separate market (Cohen & Frazzini 2008; Menzly & Ozbas 2010), a rise in cattle prices this month should predict sector-relative underperformance next month. The signal is s_t , the month's front Live Cattle return divided by its standard deviation over the 36 prior months. Each stock is hedged with its sector ETF (XLP, XLY) at a 252-day beta; the position is $w = -\text{clip}(s/2, -1, 1) \times 10\% / \hat{\sigma}_{\text{basket}}$. Tests: **Q1** same-month exposure ($\beta < 0$, $t < -2$); **Q2** next-month predictability ($\beta < 0$, $t < -2$); **Q3** positive net returns with $t > 2$ in-sample and a positive holdout Sharpe.

Timeline. H1 was committed on 2026-10-02 before any backtest; its deviation log has 21 dated entries, each recorded before the results it affects. H2 was conceived after H1 failed and committed on 2026-10-03 before any equity data was loaded; its implementation choices were logged before fitting.

2. Data, publication timing and execution

Source	Used for	Timing used in tests
Databento GLBX.MDP3 (CME Globex): definitions, final settlements, daily bars for LE, GF, ZC	Margin, prices, P&L, limit-lock test, ADV	Signal on month-end settlement t ; trade at the next day's settlement. Contract expiries from exchange definitions as published at t .
USDA NASS Cattle on Feed (feedlots 1,000+ head)	P1 outcome only	Dated by reference month (revised values); outcomes restricted to months before 2024-10.
CFTC Disaggregated COT, Live Cattle (producer/merchant positions)	Variant C filter	Known from as-of date + 6 days (+8 in holiday weeks); reports delayed by the 2013 and 2018–19 shutdowns and the 2023 ION incident held to CFTC-announced dates.
yfinance adjusted closes: TSN, TXRH, XLP, XLY; SPY	H2 returns; H1 factor check	Signal on the month-end LE settlement; trade at the next NYSE close.

Accounting. Capital is a fixed \$1M; returns are excess returns with no interest on collateral; futures positions use fractional contracts. P&L accrues within each contract (no spliced series); rolls close and reopen at one settlement and pay costs on both legs. A trade is deferred if any leg is limit-locked (official session high = low) or missing from the feed. H1 costs are one tick plus

\$2.50 per contract per side; H2 costs are 5 bps per side on stocks, 2 bps on ETFs and 0.50% a year borrow on short notional. Every result is net of costs, with a 2× cost stress.

Development verification. All H1 executions occurred one trading day after the signal, except one Variant B order deferred one extra day. Charged costs match the tick arithmetic for every trade. Variant A's gross exposure stayed below 2× capital (maximum 1.99×); Variant B is uncapped by design and reached 5.5×. H2 executed one NYSE day after each signal; stock-plus-hedge gross peaked at 1.74× and no single stock exceeded 0.43×. Known feed gaps (eight LE/GF days, two ZC days, a missing LEZ2014 final settlement) are treated as non-trading days, never forward-filled; every development output ends on or before 2024-09-30. H1 trades 2013-07-01 to 2024-09-30 (2,834 days); H2 trades 2013-08-01 to 2024-09-30 (2,810 days) from **134 monthly signals**, which are overlapping-information monthly decisions rather than independent observations.

3. H1 primary findings and trial summary

Test (pre-registered)	Estimate	NW	t	n	Result
P1: placements YoY (t+1..t+3) on z, 3 lags	$\beta = +0.0020$ per s.d.	0.42		133 month-ends	Fail (needs $\beta > 0$, $t > 2$)
P2: Variant A mean monthly net return, 5 lags	−0.045%	−0.34		135 months	Fail (in-sample)
P3 (i): sale-contract return to expiry on z, 6 lags	$\beta = -0.0087$	−0.65		127	Futures not predicted
P3 (ii): spot change on z, 6 lags	$\beta = +\$0.020/\text{lb}$	1.10		127	Spot not predicted (wrong sign)

Run (net of 1× costs unless noted)	Ann. return	Ann. vol	Sharpe	Max DD	Turnover/yr
Variant A primary, net	−0.54%	6.4%	−0.08	25.5%	6.5×
Variant A primary, gross / 2× costs	−0.39% / −0.69%	6.4%	−0.06 / −0.11	24.4% / 26.8%	6.5×
Variant A with drawdown overlay (pre-declared report)	−0.92%	5.0%	−0.18	20.7%	4.9×
Variant B, crush spread (secondary)	−1.94%	7.0%	−0.28	38.1%	23.4×
Variant C, hedging-pressure filter (secondary)	−0.18%	4.8%	−0.04	20.3%	4.4×

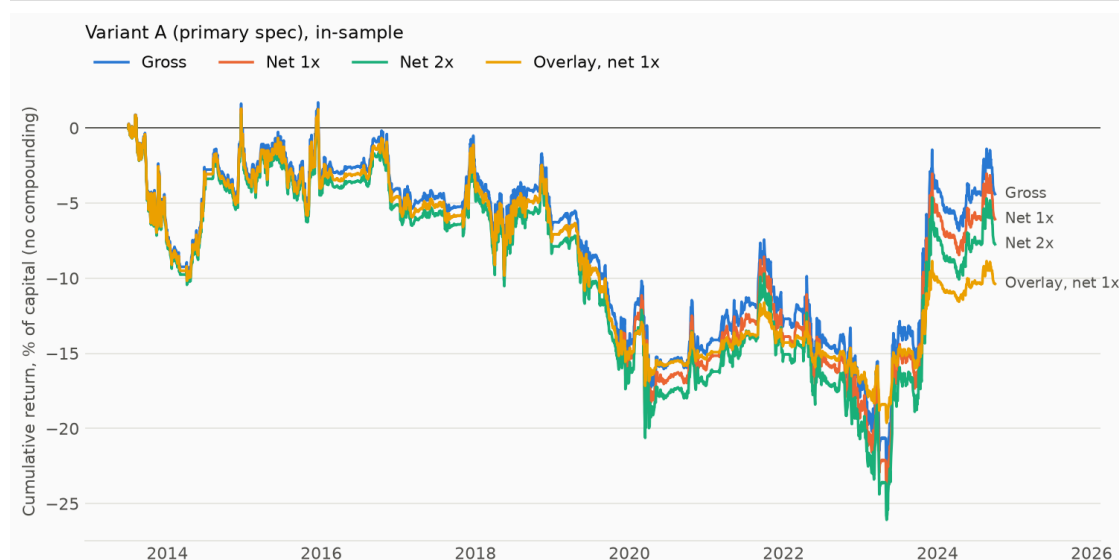


Figure 1. H1 primary (Variant A), cumulative return on fixed capital, 2013-07 to 2024-09: gross, net, net at 2× costs, and the pre-declared drawdown overlay.

Why it failed. When $z < -1$ (15 month-ends), the margin recovered by +\$153 per head over five months, almost entirely through the Live Cattle leg (+\$156). But that leg compares different contract months at t and $t+5$; P3 shows that neither the held contract's return nor the realized spot change was predictable from z . A pre-declared diagnostic finds that the calendar-seasonal component of the margin alone explains 74% of Variant A's monthly return variance over 2018–2024 (81% for C), while an LE-price-only version explains 4%. An exploratory diagnostic (not a test) found that the LE-price z alone carried a negative association with placements ($t = -3.31$), consistent with the cattle cycle; no strategy was built on it.

4. H2 primary findings and costs

Test (pre-registered, NW 2 lags)	β	NW	t	n	Result
Q1: monthly hedged basket return on same-month LE return	+0.0098	0.10		170 months	Fail: no exposure (wrong sign)
Q2: next month's hedged basket return on s_t	+0.0036	0.88		134 monthly signals	Fail (wrong sign)
Q3: mean monthly net return (in-sample)	−0.15%	−1.08		134 months	Fail

H2 primary	Ann. return	Ann. vol	Sharpe	Max DD from peak	Max loss below initial NAV
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Gross	-1.49%	6.4%	-0.23	28.7%	-26.1%
Net	-1.84%	6.4%	-0.29	30.8%	-28.9%
2× costs	-2.20%	6.4%	-0.34	33.2%	-31.6%

Costs and exposure. Turnover was 6.9× capital a year; trading costs cost 0.25% a year and borrow 0.11% a year. Average gross exposure (stocks plus hedges) was 0.42× capital, average stock exposure 0.23× and net exposure close to zero. Only 10 of the 134 monthly signals were at full conviction ($|s| \geq 2$), so the rule usually ran well below its 10% volatility target. **Concentration:** of the -16.6% cumulative gross P&L, the TXRH pair contributed -12.1% and the TSN pair -4.5%; the largest months were March 2020 (-7.8%), October 2015 (-7.1%) and December 2023 (+6.1%); excluding the three largest months, cumulative P&L is still -7.8%, with 51% of months positive. **Benchmark:** H2's benchmark is each stock's sector. Held long, the sector-hedged basket earned +0.52% a month ($t = 1.57$), so the stocks beat their sectors on average; timing them with the cattle signal lost money.

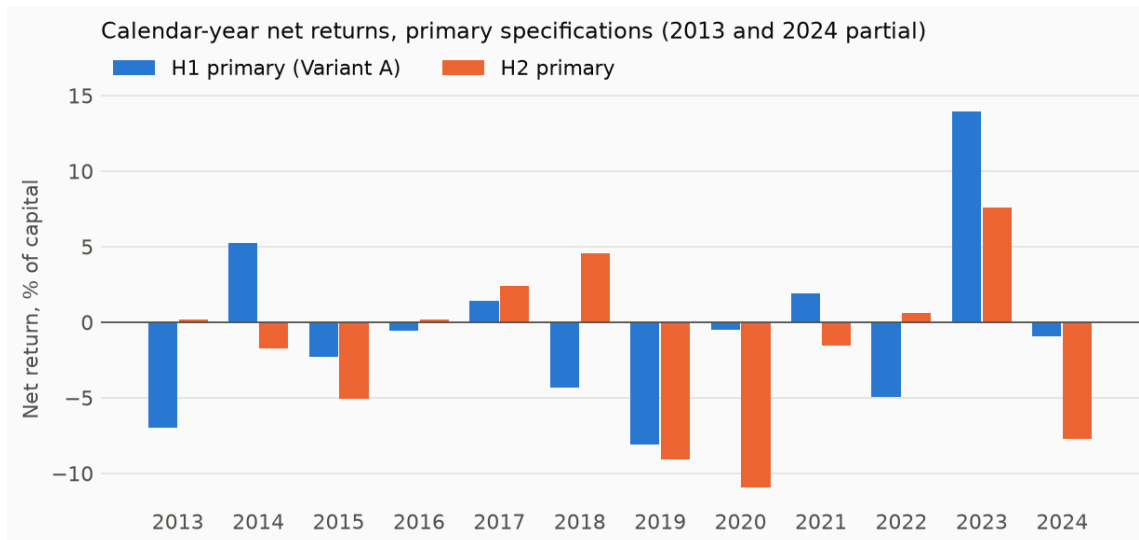


Figure 2. Calendar-year net returns of the two primary specifications (2013 and 2024 partial years).

5. Robustness, uncertainty, limitations and conclusion

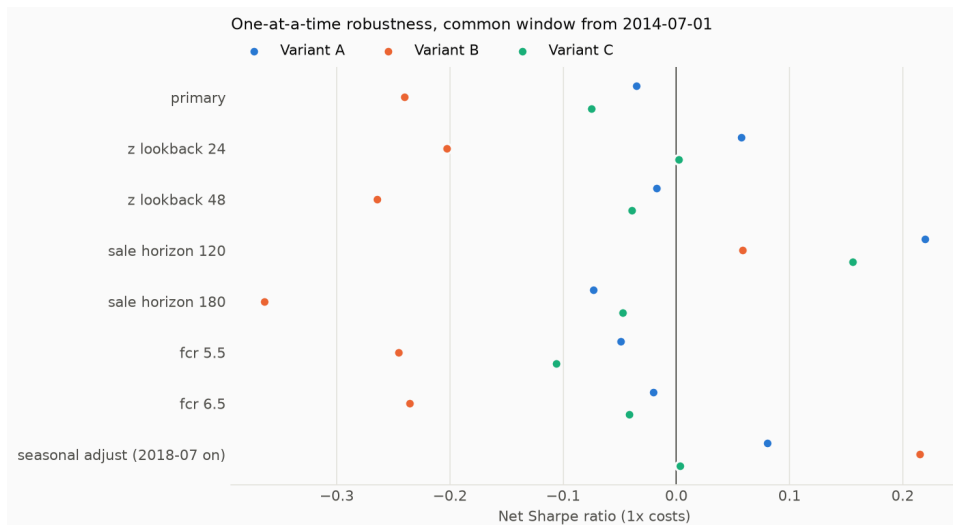


Figure 3. H1 one-at-a-time robustness (21 alternative specifications plus the 3 primaries), net Sharpe on the common window from 2014-07-01; the seasonal-adjustment specification can only start in 2018-07.

Robustness. Changing the z lookahead (24, 48 months), sale horizon (120, 180 days), feed conversion (5.5, 6.5) or seasonal adjustment moved full-sample net Sharpe ratios between -0.42 and +0.21 (up to +0.22 on the common window), with no monthly t-statistic above 0.86. The overlay and 2× costs leave the conclusion unchanged.

Deflated Sharpe ratio (25 cattle-project trials)	Net Sharpe	PSR (vs 0)	DSR
H1 primary (Variant A)	-0.08	0.39	0.09
H2 primary	-0.29	0.17	0.02
Best of the 25 trials (H1 Variant B, seasonal adjustment)	+0.21	0.70	0.40

The calculation covers the 25 distinct cattle-project trials (24 for H1, 1 for H2; expected maximum Sharpe under the null 0.31 a

year). It does not cover other experiments in the team's repository, which also holds separate GLP-1 equity, equity pairs-trading and macro-equity studies; those are disclosed here but not counted. **Other checks:** regressing Variant A on long-only cattle, 12-month cattle momentum and SPY gives alpha -1.0% a year ($t = -0.64$, $R^2 = 0.08$). Stress: Variant A lost 5.5% in the March–May 2020 packer shutdowns and 1.4% around the August 2019 Holcomb plant fire; H2 lost 9.9% in March–May 2020. Two consecutive Live Cattle limit moves (8.50 then 12.75 cents/lb) at the $2\times$ leverage cap would cost 22.9% of capital. Variant A's median position was 0.11% of 20-day volume and 1.9% initial margin-to-equity; under the square-root impact model, net Sharpe falls from -0.11 at $\$1M$ to -0.95 at $\$1B$.

Limitations. Small samples (15 month-ends with $z < -1$; 134 monthly H2 signals); feed gaps and missing final settlements in the futures data; inferred final settlements before 2017; revised placement data; current exchange margins applied historically; a two-stock H2 universe exposed to idiosyncratic events and survivorship. **Not completed:** H2's four registered alternative trials, its Fama-French factor regression and its liquidity analysis; CME exchange-set position limits for LE; a whole-pipeline perturbation test for look-ahead; an Iowa State closeout comparison of the margin's level. None of these is implied to have passed.

Conclusion. Neither pre-registered mechanism found support in the tested specifications: feedlot margins did not forecast placements or futures returns in development, and beef-exposed equities did not respond to cattle prices with a lag. H1's positive two-year holdout does not change the pre-registered verdict (Section 6). We report this as a negative result. It does not establish that no cattle-related strategy is profitable.

6. Holdout results (evaluated once)

The frozen primaries were evaluated once with `run_oos.py` at commit 58d6299. The lock was written at 2026-10-04 04:45 UTC, before any holdout result was computed (config SHA-256 ef37283b..., unchanged since development). A first attempt stopped on a Databento HTTP 502 error during the data download, before the lock was written and before any result was computed; the re-run completed. Holdout: 2024-10-01 to 2026-10-02 (504 trading days).

Holdout, net of $1\times$ costs unless noted	Ann. return	Ann. vol	Sharpe	Max DD	Monthly mean (NW t)	Holdout condition
H1 primary (Variant A)	+3.88%	6.1%	+0.63	5.5%	+0.31% (0.97)	P2: Sharpe > 0 , met
H1 primary, gross / $2\times$ costs	+3.97% / +3.79%	6.1%	+0.65 / +0.62	5.4% / 5.5%		
H2 primary	-1.03%	6.7%	-0.15	7.6%	-0.08% (-0.27)	Q3: Sharpe > 0 , not met
H2 primary, gross / $2\times$ costs	-0.51% / -1.56%	6.7%	-0.08 / -0.23	7.2% / 8.0%		
H1 Variant B (secondary)	+8.03%	9.3%	+0.86	6.5%	+0.64% (1.40)	reported, not substituted
H1 Variant C (secondary)	+2.14%	5.0%	+0.43	5.6%	+0.17% (0.80)	reported, not substituted

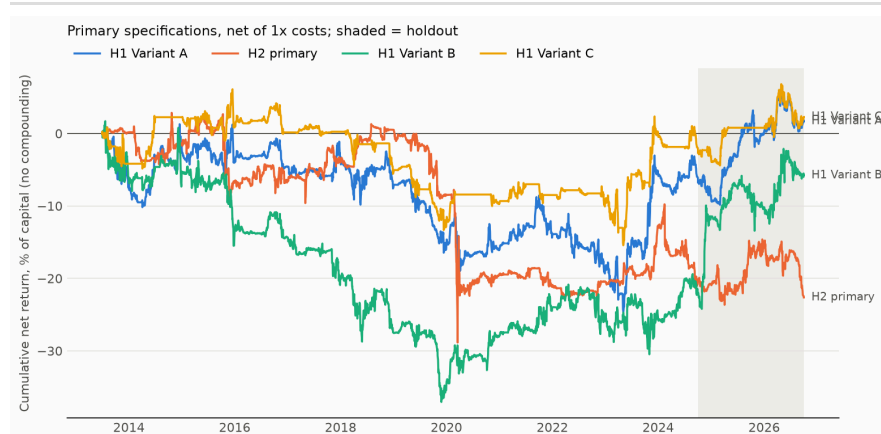


Figure 4. Cumulative net return on fixed capital, 2013 to 2026-10, holdout shaded. H1 Variant A and H2 are the frozen primaries; H1 Variants B and C are pre-registered secondary rules shown for completeness.

Interpretation. H1's primary was profitable in the holdout, and its holdout Sharpe ratio meets P2's out-of-sample condition. Two years is a short sample: the standard error of an annualized Sharpe ratio over two years is about 0.7, so +0.63 is not statistically distinguishable from zero (monthly $t = 0.97$) or from the development estimate of -0.08 . P2 required both conditions, and its in-sample condition failed; P1, the mechanism test, had failed before any trading result was seen. Under the pre-registered decision table H1 remains rejected. The holdout result is reported as observed. H2's primary lost money in the holdout and fails Q3. Variants B and C were also positive in the holdout; they are pre-registered secondary rules and are not substituted for the primary. With the drawdown overlay, B and C stayed at half size throughout the holdout (their overlaid runs' own drawdowns, carried over from development, stayed above the 7.5% restore threshold at every holdout month-end), so their overlay Sharpe ratios equal the unscaled ones.